

Music Era and Release-Market Classification

DeepMIR 2026 Fall — Homework 1

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Code, checkpoints, README: <https://github.com/chijenpeng/26-fall-DeepMIR-HW1>

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Experimental Setting

Pipeline

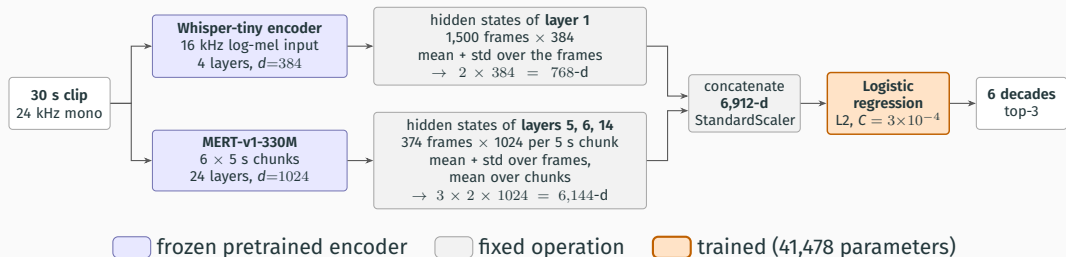
- 30 s clip, 24 kHz mono
- frozen encoder $\rightarrow$ mean + std pooling over time
- StandardScaler $\rightarrow$ L2 logistic regression
- train = fit, validation = model selection, test = prediction file only

Hardware / software

GPU	2 $\times$ NVIDIA RTX A4500 (20 GB)
CPU / RAM	Intel i9-14900K, 125 GB
OS	Ubuntu 26.04
Software	PyTorch 2.14, Transformers 4.45, scikit-learn 1.9

Results

Task 1: model



- **Whisper-tiny, layer 1:** shallow layers work best for decade; accuracy drops with depth
- **MERT-v1-330M, layers 5, 6, 14:** best three of a 25-layer sweep; adding Whisper-tiny gives +2.3 points
- **mean + std pooling:** one d -dim vector per frame $\rightarrow$ mean and std **over the frames** $\rightarrow 2 \times d$ values per layer
- Whisper: **1,500 frames** per 30 s clip. MERT: **374 frames** per 5 s chunk (its training length); statistics are taken per chunk, then averaged over the six
- Only the classifier is trained, with strong L2 (1,026 clips, 6,912 features)

Task 1: release decade

Model: frozen encoders + logistic regression

- Whisper-tiny encoder, **layer 1**
- MERT-v1-330M, **layers 5, 6, 14**
- mean + std pooling over time (6,912-d)

Model and layers **chosen by ablation:** [▶ Section 3](#)

Validation, 132 clips	Top-1	Top-3
Best model	52.3%	86.4%

Errors

- **57%** of errors land on a **neighboring decade**
- Hardest: 2000s (recall .23), absorbed by 2010s and 1990s
- Easiest: 1960s and 2010s (one neighbor each)

True	1960s	17	2	0	3	0	0
	1970s	5	9	3	2	2	1
	1980s	2	2	13	2	3	0
	1990s	1	3	2	9	4	3
	2000s	2	2	1	5	5	7
	2010s	0	2	0	0	4	16
		1960s	1970s	1980s	1990s	2000s	2010s
		Predicted					

Counts. Rows = true decade (22 clips each).

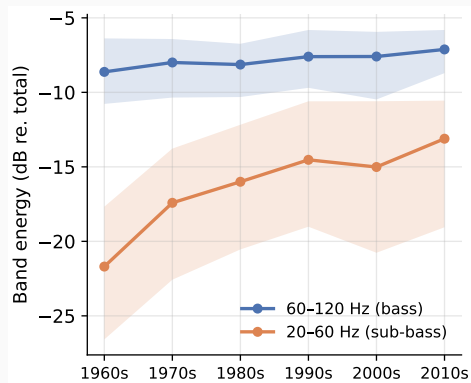
Why neighbors get confused (1): the low end

Hypothesis. Old masters were cut for vinyl, where deep bass makes the stylus jump, so they should carry less sub-bass.

Analysis. Long-term average spectrum (Welch) of the 1,026 training clips, split into two bands.

dB re. total	60s	70s	80s	90s	00s	10s
20-60 Hz	-21.7	-17.4	-16.0	-14.5	-15.0	-13.1
60-120 Hz	-8.6	-8.0	-8.1	-7.6	-7.6	-7.1

- Sub-bass rises by 8.6 dB from the 1960s to the 2010s; the 60-120 Hz band barely moves
- From the 1990s on the low end is nearly identical, so it cannot separate 1990s / 2000s / 2010s



Median and inter-quartile range, training split.

Why neighbors get confused (2): the loudness war

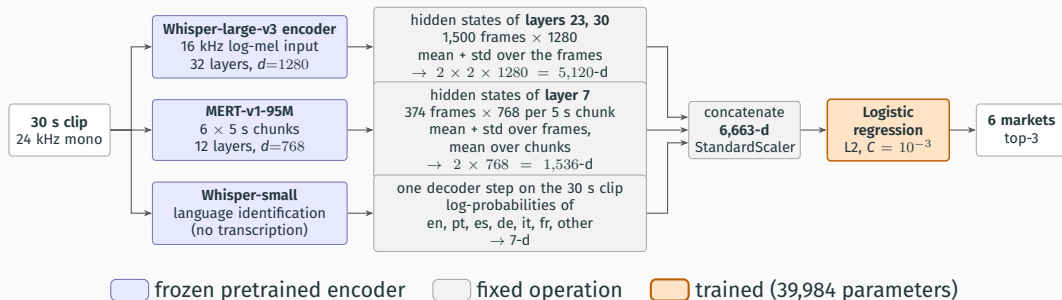
Hypothesis. From the 2000s, brick-wall limiters make masters louder and flatten peaks.

Median per decade, training clips	60s	70s	80s	90s	00s	10s
RMS level (dBFS)	-14.0	-14.6	-14.5	-13.0	-11.0	-10.6
Crest factor, peak – RMS (dB)	13.9	14.5	14.5	13.0	11.0	10.6
Dynamic range, loudest 20% blocks (dB)	10.6	11.2	11.3	10.7	8.9	8.7
Longest run pinned at the peak (samples)	5	5	6	9	22	26
Samples within 0.5 dB of the peak (%)	0.02	0.01	0.02	0.06	0.42	0.50

- 2000s and 2010s clips are 3 dB louder, have 3 dB less crest, and hold peaks at the ceiling. The two are **almost indistinguishable**, matching the 2000s ↔ 2010s confusion
- Production cues only split the data into “old” and “new”; finer distinctions must come from musical content

Caveat. The clips are 24 kHz mono, decoded from lossy Opus (YouTube). Resampling, downmixing and lossy coding reshape waveform peaks, so these numbers are trends (ANOVA $p < 10^{-30}$), not exact properties of the original masters. The two sample-level rows are the most affected.

Task 2: model



- **Whisper-large-v3, layers 23, 30**: deepest layers work best, the opposite of Task 1; this fits language being the main cue
- **MERT-v1-95M, layer 7**: +2 points on top of Whisper (57.8% $\rightarrow$ 59.8%); a music encoder adds what a speech encoder misses
- **Whisper-small language ID**: an explicit 7-d posterior; a small gain (top-3 89.2% $\rightarrow$ 90.2%) because deep Whisper layers already encode language
- Pooling as in Task 1: 1,500 Whisper frames per clip; 374 MERT frames per 5 s chunk

Task 2: release market

Model: frozen encoders + logistic regression

- Whisper-large-v3 encoder, **layers 23, 30**
- MERT-v1-95M, **layer 7**
- Whisper-small language-ID posteriors

Model and layers **chosen by ablation:** [▶ Section 3](#)

Validation, 102 clips	Top-1	Top-3
Best model	60.8%	90.2%

Errors

- **US ↔ UK** is the largest confusion (9 clips)
- **UK is over-predicted:** 23 called UK, only 9 correct
- Brazil (.88) and Spain (.82) are rarely missed
- Germany, Italy: 12 of their 21 errors go to US / UK

True	US	11	4	0	0	1	1
	UK	5	9	0	0	2	1
	Brazil	0	0	15	0	0	2
	Spain	0	2	0	14	1	0
	Germany	3	4	1	2	6	1
	Italy	1	4	0	2	3	7
		US	UK	Brazil	Spain	Germany	Italy
		Predicted					

Counts. Rows = true market (17 clips each).

Why English-language markets get confused: language

The dataset has **no language label**. We estimate it with Whisper-small **language identification** on the 798 training clips, on the mixture and on Demucs-separated vocal stems.

Market	Home language	Home mixture	Detected as English	
			mixture	vocals
US	English	92%	–	–
UK	English	100%	–	–
Brazil	Portuguese	87%	11%	12%
Spain	Spanish	57%	39%	41%
Germany	German	17%	83%	82%
Italy	Italian	41%	55%	56%

Share of training clips whose most likely language (among en, pt, es, de, it) is the home language or English.

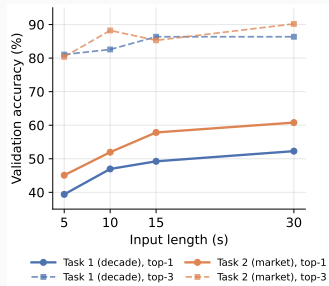
- Most German releases and over half of the Italian ones are **sung in English**, so their vocals carry no market cue
- Vocal stems give the same answer: the accompaniment is not what misleads the detector
- Language separates Brazil and Spain. The English-language group has to be split by genre and production style, a much weaker cue
- A release market is where the record was issued, not the artist's nationality or language

Optional Experiments

Input length: 5 / 10 / 15 / 30 s

Best models on the first k seconds only; features re-extracted and classifier re-trained per length.

Length	Task 1		Task 2	
	Top-1	Top-3	Top-1	Top-3
5 s	39.4	81.1	45.1	80.4
10 s	47.0	82.6	52.0	88.2
15 s	49.2	86.4	57.8	85.3
30 s	52.3	86.4	60.8	90.2



- Top-1 rises monotonically on both tasks and has **not saturated** at 30 s; Task 1 top-3 is flat after 15 s
- Task 2 gains more (+15.7 vs. +12.9 points)

Multiple excerpts per recording

Each 30 s clip is cut into six 5 s chunks. **Chunk-level:** every chunk is its own training sample ($6\times$ more data) and chunk probabilities are averaged at test time. **Clip-level:** chunk features are averaged first (our default).

Task	Features	Clip-level		Chunk-level	
		Top-1	Top-3	Top-1	Top-3
1	MERT-v1-95M, layers 6, 8, 10	48.5	81.8	47.7	84.8
1	MERT-v1-330M, layers 5, 6, 14	50.0	87.9	50.0	86.4
2	MERT-v1-95M, layer 7	42.2	72.5	45.1	76.5
2	MERT-v1-330M, layers 8, 9, 12	42.2	69.6	44.1	76.5

- Task 1: no gain. Six chunks of one recording are highly correlated, so they add little new information for a linear model
- Task 2: a small gain (+2 to +3 points top-1, +4 to +7 top-3), about 2 to 3 validation clips
- We keep clip-level features for the final models: same accuracy, six times fewer training rows

MERT features only, not the full final models: Whisper features were extracted on whole 30 s clips.

Task 1 as year regression

Decades are ordered, so we test a formulation that uses the order.

Features (same as the final Task 1 model):

Whisper-tiny encoder layer 1 + MERT-v1-330M layers 5, 6, 14, mean + std pooling, 6,912-d.

Classification: L2 logistic regression, $C = 3 \times 10^{-4}$.

Regression: ridge regression, $\alpha = 10^4$, on a decade index (0 = 1960s, 1 = 1970s, ..., 5 = 2010s), rounded to the nearest decade; top-3 = the three nearest decades.

Formulation	Top-1	Top-3
Logistic regression (final model)	52.3	86.4
Ridge regression	31.8	77.3

Mean absolute error of the regression: 0.98 decades.

True	1960s	3	10	6	3	0	0
	1970s	1	8	8	4	1	0
	1980s	0	2	10	9	1	0
	1990s	0	0	5	9	7	1
	2000s	0	1	5	6	9	1
	2010s	0	1	1	7	10	3
		1960s	1970s	1980s	1990s	2000s	2010s
		Predicted					

Ridge regression. Counts; rows = true decade (22 clips each).

Task 1 as year regression: why it fails

Both rows use the final Task 1 features (Whisper-tiny layer 1 + MERT-v1-330M layers 5, 6, 14); only the predictor differs.

Number of validation clips each formulation assigns to each decade:

Clips predicted as	1960s	1970s	1980s	1990s	2000s	2010s
True count	22	22	22	22	22	22
Logistic regression (final model)	27	20	19	21	18	27
Ridge regression	4	22	35	38	28	5

- **Regression is much worse** (−20.5 points top-1): predictions collapse toward the middle, only 4 clips are called 1960s and 5 are called 2010s
- Tuning does not fix it: with a ridge penalty from 10^2 to 10^6 the best top-1 is 32%; stronger regularization worsens the collapse
- The classifier already learns the order by itself (see the t-SNE slide), so nothing is gained from imposing it

Task 2: mixture vs. vocals vs. accompaniment

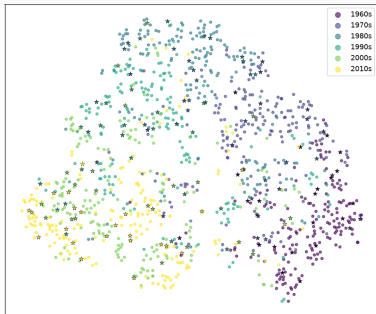
Demucs (htdemucs) separates each clip into vocals and accompaniment. The **full final pipeline** (Whisper-large-v3 layers 23, 30 + MERT-v1-95M layer 7 + language ID) is applied to each input.

Input	Top-1	Top-3
Mixture (final model)	60.8	90.2
Vocals only	60.8	87.3
Accompaniment only	35.3	70.6
Mixture + vocals	60.8	90.2
Mixture + vocals + accompaniment	61.8	90.2

Recall	US	UK	BR	ES	DE	IT
Mixture	.65	.53	.88	.82	.35	.41
Vocals	.59	.76	.82	.76	.35	.35
Accompaniment	.41	.59	.59	.06	.18	.29

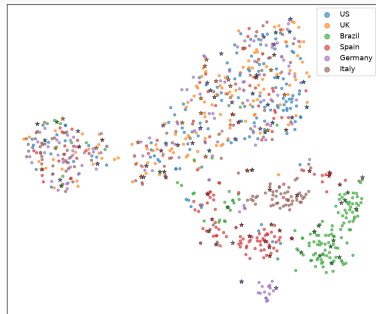
- **Market information is in the vocals:** vocals alone match the mixture; removing them costs 25 points
- Without vocals Spain falls to 6% recall, and language ID detects no Portuguese, Spanish, German or Italian at all
- Accompaniment alone stays above chance (16.7%): a weak market signal in the instrumental
- Separation does not help: all three inputs gain one clip at three times the feature size, and US ↔ UK confusions stay at 8 to 9. The final model uses the mixture

t-SNE of the best models



Task 1: 6-d logit space of the classifier

- Task 1: decades line up along **one arc from the 1960s to the 2010s**, although the classifier was never told they are ordered; 2000s and 2010s overlap
- Task 2: Brazil, Spain and Italy form clusters **before any training**; US, UK and most of Germany share one English-language cloud



Task 2: raw feature space, no classifier

Dots = train, stars = validation. Train points in the logit space are fitted; only the stars show generalization.

Confusion patterns and dataset bias

The detailed analyses are in the Results section. Summary, plus two additional findings:

	Task 1: decade	Task 2: market
Confusion pattern	57% of errors on a neighboring decade; 2000s ↔ 2010s is the worst pair	75% of errors among US, UK, Germany, Italy; UK is over-predicted
Bias in the data	production changes with the era: sub-bass +8.6 dB, crest factor −3 dB after 2000	English-language releases: 83% of Germany, 55% of Italy
What the model keys on	mastering loudness and bandwidth as well as musical content	the language of the vocals rather than the market itself

- **Hand-crafted production features** (24-d: band energies, crest factor, limiter statistics) reach 40.9% on Task 1 but only **21.6% on Task 2** (chance 16.7%): within one decade, mastering does not depend on the market
- **High-frequency energy** (8–12 kHz) is 4 to 5 dB lower in the 1960s and 1970s than later; the clips are band-limited to 12 kHz, so anything above that is unavailable

Ablation

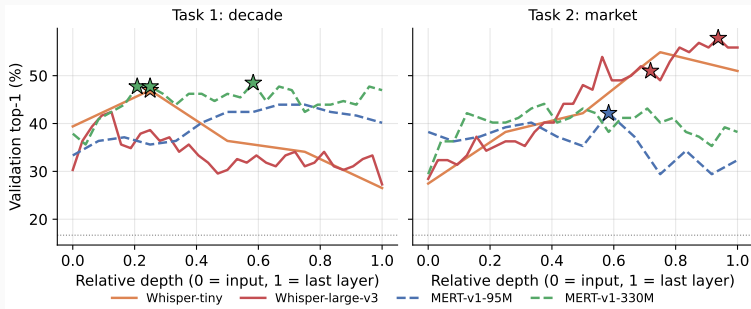
Ablation (1): which features?

Task 1: decade	Top-1	Top-3
Hand-crafted production (24-d)	40.9	85.6
MERT-v1-95M, best layer (8)	43.9	84.8
MERT-v1-95M, layers 6, 8, 10	48.5	81.8
Whisper-tiny, layer 1	47.0	81.8
MERT-v1-330M, layers 5, 6, 14	50.0	87.9
+ Whisper-tiny layer 1 (final)	52.3	86.4
+ hand-crafted	51.5	86.4

Task 2: market	Top-1	Top-3
Hand-crafted production (24-d)	21.6	58.8
MERT-v1-95M, layer 7	42.2	72.5
MERT-v1-330M, layer 9	45.1	72.5
Whisper-small language ID (7-d)	53.9	79.4
Whisper-large-v3, layer 30	57.8	84.3
+ MERT-v1-95M layer 7	59.8	89.2
+ language ID	59.8	90.2
+ layer 23 (final)	60.8	90.2

- **Task 1:** a music encoder beats a speech encoder, and the two are complementary (+2.3). Hand-crafted features add nothing once MERT is present
- **Task 2:** the order is reversed. A speech encoder beats explicit language ID, which beats a music encoder; production features are at chance (16.7%)
- All rows: mean + std pooling, StandardScaler, L2 logistic regression; C chosen on validation from $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$

Ablation (2): which layer?



- One logistic regression per layer; stars mark the layers used in the final models
- **Task 1:** Whisper is best at its **first** layers (47.0% at layer 1 of Whisper-tiny) and falls to 27% at the last. MERT peaks in the middle
- **Task 2:** Whisper **improves with depth**, from 32% at layer 1 to 58% at layer 30 of Whisper-large-v3
- Same encoder, opposite trends: decade cues sit in shallow acoustic layers, market cues in deep layers

Ablation (3): encoder size, pooling and classifier

Encoder size: top-1 of the best frozen layer

Encoder	Layers	Task 1 (layer)	Task 2 (layer)
Whisper-tiny	4	47.0 (1)	54.9 (3)
Whisper-base	6	37.1 (1)	51.0 (4)
Whisper-small	12	41.7 (2)	56.9 (12)
Whisper-medium	24	43.2 (4)	55.9 (13)
Whisper-large-v3	32	42.4 (4)	57.8 (30)
MERT-v1-95M	12	43.9 (8)	42.2 (7)
MERT-v1-330M	24	48.5 (14)	44.1 (9)

Pooling and classifier

Task 1, MERT-v1-330M	Top-1	Top-3
mean + std, log. reg.	50.0	87.9
mean only, log. reg.	46.2	87.1
mean + std, RBF SVM	43.9	82.6

- Size helps in opposite ways: for Task 1 the **smallest** Whisper is best, for Task 2 the **largest**. A larger MERT helps both
- The std over time is worth +3.8 points; logistic regression beats an RBF SVM on these features
- **Model selection:** encoder, layers, C and epoch are chosen by validation top-1. Over 100 configurations were compared on 132 and 102 clips, so validation scores are optimistic estimates of test accuracy

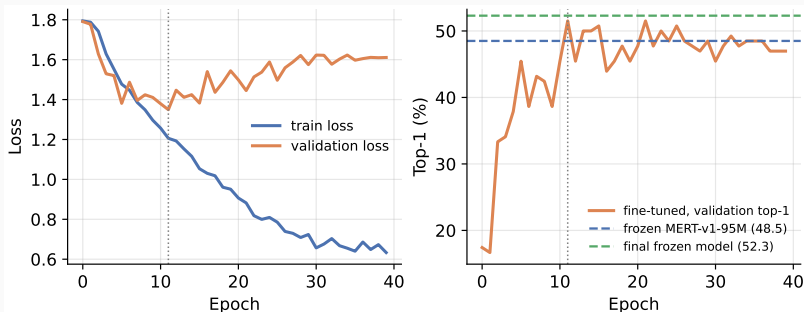
Ablation (4): frozen features vs. fine-tuning

Encoder	Params	Epochs	Batch	Task 1 top-1		Task 2 top-1	
				fine-tuned (best epoch)	frozen	fine-tuned (best epoch)	frozen
MERT-v1-95M	95 M	8	16	46.2 (6)	48.5	39.2 (6)	42.2
MERT-v1-95M	95 M	40	16	51.5 (11)	48.5	46.1 (16)	42.2
MERT-v1-330M	330 M	40	16	48.5 (20)	50.0	44.1 (26)	45.1
Whisper-tiny	8 M	20	8	34.8 (4)	47.0	50.0 (3)	54.9
Whisper-base	20 M	20	8	31.8 (3)	37.1	54.9 (13)	51.0
Whisper-small	88 M	20	8	34.8 (4)	41.7	55.9 (2)	56.9
Whisper-medium	307 M	20	2	41.7 (12)	43.2	57.8 (4)	55.9
Final frozen combination (logistic regression)					52.3		60.8

Setup. Linear head on a learned weighted sum of all layers, mean pooling. AdamW; encoder lr 2×10^{-5} (Whisper 10^{-5}), head lr 10^{-3} ; weight decay 0.01; one-cycle schedule; label smoothing 0.1; gradient clipping 1.0; bf16. MERT: random 5 s crops. Whisper: full 30 s clip. “Frozen” = best layer(s) of the same encoder + logistic regression.

- Fine-tuning beats the frozen features of the same encoder in 4 of 14 runs and **never beats the final frozen combination**. 8 epochs were too few for MERT, but beyond about 12 epochs it only overfits (next slide)

Ablation (5): fine-tuning overfits



MERT-v1-95M fine-tuned on Task 1 for 40 epochs; a checkpoint was saved after every epoch.

- Validation loss is lowest at **epoch 11** and rises afterwards while training loss falls to 0.63: the model memorizes the 1,026 training clips
- The Whisper runs overfit faster (validation loss minimum at epochs 2 to 5); their learned layer weights concentrate on the *last* layer, the worst one for Task 1
- With 130 to 170 clips per class, a frozen encoder with a strongly regularized linear classifier is as accurate, more stable, and far cheaper

Audio Language Model

Qwen2-Audio, zero-shot: setup

Model	Qwen2-Audio-7B-Instruct, no training
Inference	bf16, one RTX A4500 (20 GB), batch 1, 2.2 s per clip
Input Data	full 30 s clip, resampled to 16 kHz all validation and test clips: 264 (Task 1), 204 (Task 2)
Decoding	greedy, at most 16 new tokens

Two outputs per clip

- **Free generation**, parsed with a regular expression (“United States” → US, etc.). No label found = *invalid*, counted as wrong
- **Log-probability ranking**: each of the six labels is scored by the log-probability of its tokens given the prompt. Every clip gets a full ranking: top-3 is defined and **no output is invalid**

Prompt “naive” (the audio precedes the text)

Task 1. *This is a 30-second excerpt of a music recording released in the United States. Our goal is to identify the decade in which the recording was released. Which decade is it? Answer with exactly one of: 1960s, 1970s, 1980s, 1990s, 2000s, 2010s.*

Task 2. *This is a 30-second excerpt of a music recording released in the 1980s. Our goal is to identify the country (release market) in which this record was issued. Which market is it? Answer with exactly one of: US, UK, Brazil, Spain, Germany, Italy.*

Qwen2-Audio, zero-shot: results

Validation, naive prompt	Task 1	Task 2
Top-1, log-prob ranking	34.1	52.9
Top-3, log-prob ranking	69.7	73.5
Top-1, free generation	31.1	47.1
Invalid free outputs	1.5%	1.0%
Our model, top-1	52.3	60.8
Our model, top-3	86.4	90.2

True	1960s	14	8	0	0	0	0
	1970s	9	13	0	0	0	0
	1980s	3	16	3	0	0	0
	1990s	2	14	4	0	1	1
	2000s	0	14	1	0	0	7
	2010s	1	6	0	0	0	15
		1960s	1970s	1980s	1990s	2000s	2010s

Task 1

True	US	16	1	0	0	0	0
	UK	8	9	0	0	0	0
	Brazil	1	1	15	0	0	0
	Spain	2	2	2	11	0	0
	Germany	8	4	1	3	1	0
	Italy	7	2	1	5	0	2
		US	UK	Brazil	Spain	Germany	Italy

Task 2

- Task 1: **55%** of the free answers are “1970s” and none is “1990s”; the 1980s to 2000s are almost all called 1970s
- Task 2: Brazil and Spain are recognized, but English-language songs are called **US**: Germany 1 of 17 and Italy 2 of 17 correct
- The same weakness as our language feature: the model hears the language, not the market

Confusion matrices: naive prompt, log-probability ranking, counts, rows = true class.

Qwen2-Audio: three prompt designs

Task	Prompt	Words	Log-prob ranking		Free generation		
			Top-1	Top-3	Top-1	Invalid	Most frequent free answer
1	naive	42	34.1	69.7	31.1	1.5%	1970s, 55% of clips
1	cues	74	26.5	62.1	24.2	1.5%	1970s, 74%
1	production	271	17.4	58.3	18.2	6.1%	2010s, 92%
2	naive	43	52.9	73.5	47.1	1.0%	US, 54%
2	cues	78	22.5	61.8	28.4	11.8%	US, 47%

cues: names what to listen for (production, instrumentation; for Task 2 the language of the lyrics) and states that all classes are equally likely. **production:** a 271-word listening guide from our own analysis (sub-bass and vinyl, tape hiss, the loudness war). Full texts are in the code.

- **The longer the prompt, the worse the result:** the text prior overrides the audio. “production” ends with a description of the 2010s, and the model answers 2010s for 92% of clips
- Task 2 “cues” contains “Portuguese suggests Brazil”: in the log-prob mode 96 of 102 clips then rank Brazil first. Label words in a prompt bias token scoring
- Telling the model what to listen for does not help: the language model only receives the last Whisper layer, where decade cues are weakest (Ablation 2)

Disclosure

Tool: Claude Code with the Claude Fable 5.1 model (Anthropic).

Done with AI assistance

- Code implementation: feature extraction, training, evaluation, figures
- Running the experiments on the server and keeping the experiment log
- Preparing these slides

My own

- The ideas and hypotheses: sub-bass and vinyl mastering, the loudness war and dynamics, language as the market cue
- The decisions: which experiments to run, how to read them, which models to submit

Every number in this report comes from running the code in the repository, and I reviewed the results and the text. The assistant also proposed implementation options, such as candidate pretrained encoders, from which I selected.

References

References (1): pretrained models and data

Pretrained models

- **MERT** (m-a-p/MERT-v1-95M, MERT-v1-330M): Y. Li et al., “MERT: Acoustic Music Understanding Model with Large-Scale Self-supervised Training,” ICLR 2024. arXiv:2306.00107
- **Whisper** (openai/whisper-tiny, base, small, medium, large-v3): A. Radford et al., “Robust Speech Recognition via Large-Scale Weak Supervision,” ICML 2023. arXiv:2212.04356
- **Qwen2-Audio** (Qwen/Qwen2-Audio-7B-Instruct): Y. Chu et al., “Qwen2-Audio Technical Report,” 2024. arXiv:2407.10759
- **Demucs** (htdemucs): S. Rouard, F. Massa, A. Défossez, “Hybrid Transformers for Music Source Separation,” ICASSP 2023. arXiv:2211.08553

Dataset

- **Discogs-VI** (source of both homework datasets): R. O. Araz, X. Serra, D. Bogdanov, “Discogs-VI: A Musical Version Identification Dataset Based on Public Editorial Metadata,” ISMIR 2024. arXiv:2410.17400

References (2): software and methods

Public codebases

- PyTorch: A. Paszke et al., NeurIPS 2019. torchaudio: Y.-Y. Yang et al., ICASSP 2022
- Hugging Face Transformers: T. Wolf et al., EMNLP 2020 (System Demonstrations)
- scikit-learn: F. Pedregosa et al., JMLR 2011. librosa: B. McFee et al., SciPy 2015
- nnAudio (CQT front end required by MERT): K. W. Cheuk et al., IEEE Access 2020
- Demucs code: <https://github.com/facebookresearch/demucs>
- NumPy, SciPy, pandas, matplotlib, seaborn, soundfile, tqdm, joblib

Methods

- t-SNE: L. van der Maaten, G. Hinton, JMLR 2008. Welch spectral estimate: P. Welch, IEEE Trans. Audio Electroacoust., 1967
- Fine-tuning: AdamW (Loshchilov & Hutter, ICLR 2019), one-cycle schedule (Smith & Topin, 2019), label smoothing (Szegedy et al., CVPR 2016), SpecAugment time masking (Park et al., Interspeech 2019)

Tool: Claude Code, Claude Fable 5.1 (Anthropic); see the Disclosure slide.